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Research paper

N100 as a response prediction biomarker for accelerated 1 Hz right DLPFC-rTMS in major depression

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ABSTRACT

Background and objective: Repetitive transcranial magnetic stimulation (rTMS) is a safe and effective treatment for major depressive disorder (MDD); however, this treatment currently lacks reliable biomarkers of treatment response. TMS-evoked potentials (TEPs), measured using TMS-electroencephalography (TMS-EEG), have been suggested as potential biomarker candidates, with the N100 peak being one of the most promising. This study investigated the association between baseline N100 amplitude and 1 Hz right dorsolateral prefrontal cortex (R-DLPFC) accelerated rTMS (arTMS) treatment in MDD.

Methods: Baseline TMS-EEG sessions were performed for 23 MDD patients. All patients then underwent 40 sessions of 1 Hz R-DLPFC (F4) arTMS over 5 days and a follow-up TMS-EEG session one week after the end of theses arTMS sessions.

Results: Baseline N100 amplitude at F4 showed a strong positive association ($p < .001$) with treatment outcome. The association between the change in N100 amplitude (baseline to follow-up) and treatment outcome did not remain significant after Bonferroni correction ($p = .06$, corrected; $p = .03$, uncorrected). Furthermore, treatment responders had a significantly larger mean baseline F4 TEP amplitude during the N100 time frame compared to non-responders ($p < .001$). Topographically, after Bonferroni correction, F4 is the only electrode at which its baseline N100 amplitude showed a significant positive association ($p < .001$) with treatment outcome.

Limitations: Lack of control group and auditory masking.

Conclusion: Baseline N100 amplitude showed a strong association with treatment outcome and thus demonstrated great potential to be utilized as a cost-effective and widely adoptable biomarker of rTMS treatment in MDD.

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1. Introduction

Major depressive disorder (MDD) is now the foremost cause of disability worldwide (Friedrich, 2017). Repetitive transcranial magnetic stimulation (rTMS) has entered increasingly widespread use as an effective evidence-based treatment for medication-resistant MDD (Brunoni et al., 2017; Lefaucheur et al., 2020; Milev et al., 2016a). A biomarker of rTMS treatment response in MDD, however, has yet to be determined. Although clinical features have shown some group-level predictive power, it has been difficult to demonstrate reliable individual-level predictive power for any biomarker to date (Simon and Perlis, 2010; Bagby et al., 2002). The ideal biomarker would, in addition to having high individual-level predictive accuracy, be cost-effective and simple to acquire in real-world clinical settings.

Since rTMS is focused on the level of the cortex, it is plausible that the outcome of rTMS treatment is related to how the cortical neurons react to the stimulation. This cortical reactivity can reflect both functional and anatomical differences across patients that results in varying rTMS treatment outcomes in MDD. To measure cortical reactivity, electroencephalography (EEG) can be combined with rTMS to examine changes in neuronal dynamics in response to electrical stimulation on a millisecond time scale. More specifically, TMS pulses can induce depolarization of local or distal neuronal networks, which in turn results in variations in the summation of pyramidal neuron and interneuron post-synaptic potentials that can be recorded by EEG electrodes as voltage fluctuations. When time locked to TMS pulses, these voltage deflections are known as TMS-evoked potentials (TEPs). Current understanding suggests that TEPs are unlikely to be the result of non-neurological artefacts, as TEPs are only observed following TMS stimulation of functionally preserved cortical regions, and are absent when stimulation targets lesioned cortical regions, despite of intact scalp and ocular muscles and nerves (Gosseries et al., 2015). TEP features such as peak amplitude and latency may thus reflect local cortical reactivity and have the potential to predict rTMS treatment outcome (Tremblay et al., 2019).

To produce the most generalizable results, the selection of TEP peaks as candidates of treatment response biomarkers should be done with careful a priori consideration. Amongst the various TEP peaks, the N100, a negative deflection closest to 100 ms after the TMS pulse, has been suggested to have a more substantial role in MDD (Arns and Voetterl, 2021) as well as one of the best test-retest reliability at the DLPFC (Kerwin et al., 2018). Potentially contributing to its great test-retest reliability, another major advantage of N100 is that, compared to earlier peaks, N100 is less prone to contamination from TMS muscle artefacts (lasting around 50 ms post TMS pulse) and decay artefacts (peaking before 50 ms post TMS pulse) (c.f. supplementary material of reference (Rogasch et al., 2014)). Thus, N100 was selected a priori as the TEP peak of interest for this study.

Although preliminary, several studies have provided insights as to the mechanism underlying TEPs and current evidence suggest that gamma-aminobutyric acid (GABA), the most prominent inhibitory neurotransmitter of the central nervous system, has an important role in MDD (Gold et al., 1980; Korpi and Sinkkonen, 2006; Kralic et al., 2002) as well as in the modulation of TEP peak amplitudes (Premoli et al., 2014; Darmani et al., 2016; Bettler et al., 2004). Significant GABA deficits (Gold et al., 1980; Hasler et al., 2007; Price et al., 2009; Schür et al., 2016) along with substantial decreases in DLPFC GABAergic interneuron size and density (Rajkowska et al., 2007) have been found in MDD patients, and rTMS responders have been shown to have a larger increase in GABA levels compared to non-responders (Dubin et al., 2016; Levitt et al., 2019). By examining the effect of various GABA receptor agonists and antagonists on the amplitude of TEP peaks, pharmacological studies have reported that the amplitude of N45, an early TEP peak, can be modulated through the GABA-A receptors, which mediates fast inhibitory processes (Premoli et al., 2014; Darmani et al., 2016), while the amplitude of N100, a later TEP peak, is decreased by GABA-A receptor agonists alprazolam and diazepam, and increased by GABA-B receptors agonist baclofen (Premoli et al., 2014). Therefore, the current understanding of TEPs suggest that the amplitude of N100 is associated with GABA-B receptor (GABABR) mediated slow inhibitory processes involving the activation of G proteins by second messengers (Premoli et al., 2014; Bettler et al., 2004) and this process can potentially be influenced by rTMS (Dubin et al., 2016; Levitt et al., 2019).

The association between N100 amplitude and treatment outcome in MDD has been reported in several studies. Using the auditory oddball ERP paradigm, Spronk and colleagues reported a decrease in N100 amplitude following rTMS treatment in MDD (Spronk et al., 2008). In a later study by the same group, again using the auditory oddball ERP paradigm, demonstrated a positive association between baseline N100 amplitude and antidepressant treatment outcome for MDD, suggesting that MDD patients with more negative N100 amplitudes at baseline will improve more on antidepressant medications (Spronk et al., 2011). In terms of TMS-EEG studies, in line with the results from the auditory oddball paradigm studies, Sun and colleagues demonstrated that a greater N100 amplitude at baseline led to a greater suicidality improvement after magnetic seizure therapy (MST) in MDD patients (Sun et al., 2016). In 2021, Voineskos and colleagues showed that the change in TEP N100 peak amplitude from baseline to post-treatment was positively associated with high frequency (HF) 10Hz arTMS treatment outcome in MDD, suggesting that a more positive change in N100 amplitude post rTMS led to larger improvements in MDD symptoms (Voineskos et al., 2021).

The primary aim of this study is to investigate the association between baseline N100 amplitude and low frequency (LF) arTMS outcome in MDD to evaluate the use of baseline N100 amplitude as a treatment outcome biomarker. To the best of our knowledge, our study is the first to examine this association in 1 Hz LF-rTMS instead of HF-rTMS. Increasing reports on the often-overlooked LF-rTMS is crucial, since LF-rTMS has several important advantages over HF-rTMS (Kaur et al., 2019) and the singular potential for use in home settings that would dramatically improve rTMS accessibility in a time when MDD is becoming increasingly prevalent (Miron et al., 2020). Since the mechanistic differences between 1 Hz and HF-rTMS are still under investigation, we hypothesized that the association between baseline F4 N100 amplitude and treatment outcome for our study would agree with the previously suggested positive association between baseline N100 amplitude and MST treatment outcome (Sun et al., 2016).

2. Methods

2.1. Study design and participants

23 MDD participants, 13 female and 10 male, between the ages of 26–61 were included in this single-arm, prospective, open-label feasibility study. Participants were included if they met criteria for MDD diagnosis on the Mini International Neuropsychiatric Interview, and have maintained a stable medication regimen from 4 weeks prior to the start of the study and could maintain a stable medication regimen throughout the study. Please see supplementary materials for the full inclusion/exclusion criteria.

2.2. EEG data acquisition

EEG acquisitions were performed at the baseline session, which occurred within one week prior to the arTMS treatment sessions, and at the follow-up session, which occurred within one week after the arTMS treatment sessions. EEG was acquired using a 16-lead, 512 Hz EEG system (EasyCap, Brain Products, Munich, Germany). The FCz and AFz electrodes on the EEG cap were used as reference and ground electrodes, respectively. Stable impedances of $<5k\Omega$ were acquired prior to the start of the recording session and monitored throughout the recording. Participants first underwent a 5-minute pre-stimulation resting state session, then a 10-minute 1 Hz R-DLPFC rTMS EEG session (1 single train, 1

second inter-pulse interval, 600 pulses per session, 120 % of resting motor threshold), followed by another 5-minute post stimulation resting state session. For the rTMS EEG session, rTMS was delivered with a MagPro R20 stimulator equipped with a MC-B70 coil (MagVenture, Farum, Denmark). The coil was placed over the F4 electrode, which most closely represents the location of R-DLPFC determined according to a previously published heuristic using scalp measurements (Mir-Moghtadaei et al., 2017). All recordings were conducted from 9 am to 12 pm during the day.

2.3. arTMS treatment procedures

arTMS treatment sessions took place within one week after the baseline session. rTMS was delivered with a MagPro R20 stimulator equipped with a MC-B70 coil (MagVenture, Farum, Denmark). Resting motor threshold (rMT) was determined according to standard techniques (McClintock et al., 2017). Treatment consisted of an arTMS course of 8 hourly sessions per day over the 5 consecutive weekdays from Monday to Friday, totaling 40 sessions. Each rTMS session consisted of low-frequency 1 Hz stimulation delivered over a 10 min period (1 single train, 600 pulses per session, 50-minute inter-session interval) at 120 % of rMT over the right dorsolateral prefrontal cortex (R-DLPFC). R-DLPFC was located according to a previously published heuristic using scalp measurements (Mir-Moghtadaei et al., 2017).

2.4. Clinical outcome

Clinical treatment outcome was measured using percent improvement in Beck Depression Inventory-II (BDI-II) (Beck et al., 1961) scores, which was calculated using the baseline and second follow-up (four weeks after the arTMS treatment session) BDI-II scores.

2.5. EEG data analysis

EEG data was preprocessed and analyzed using MNE (Gramfort et al., 2013). The first 3 min of the rTMS EEG recording was included in the analysis to isolate the time period when the patient is least accustomed to the stimulation and is most mentally alert, a decision that was made a priori. Our previous unpublished ECG analysis on the same dataset also showed that the effect of rTMS on heart rate was most significant during the first 3 min of stimulation. This 3-minute rTMS EEG recording segment includes 180 pulses, which exceeds the number of total pulses administered in previous studies (Rogasch et al., 2014; Sun et al., 2016).

Preprocessing steps were based on the previously published methods by Rogasch et al. (Rogasch et al., 2014). To remove artefacts generated by the rTMS pulse, all data from 10 ms before the pulse to 6 ms after the pulse were excluded from the analysis. A zero-phase FIR 1 Hz highpass filter and a zero-phase 60 Hz notch filter were applied. Data was epoched from 300 ms pre-pulse to 500 ms post-pulse. Examination for bad trials and channels was done manually to remove data with clearly visible contaminants such as noisy electrodes and patient movements; bad trials were removed while bad channels were interpolated. Two rounds of fast independent component analysis (FastICAs) were performed to remove artefactual components according to features presented in Rogasch et al. (Rogasch et al., 2014). The first round was used to remove remaining rTMS pulse artefacts, after which a zero-phase FIR 80 Hz lowpass filter was applied, then a second round of ICA was used to remove other artefacts including decay artefacts, eye blinks, eye movements, muscle artefacts, auditory and noise-related artefacts, and electrode artefacts. Lastly, data was baseline corrected to the period from 300 to 100 ms before the onset of the rTMS pulse and re-referenced to the all-channel average.

TEP was obtained for each electrode by averaging all trials. N100 amplitude was determined by finding the most negative peak between 80 and 130 ms after the rTMS pulse onset.

2.6. Statistical analysis

All statistical analyses were performed in R (version 4.0.4) (R Core Team, 2021). Linear regression was used to investigate the association between N100 amplitude during the baseline session and treatment outcome. This was first completed for the F4 electrode (Fig. 1), which is the site of stimulation and thus the region of interest, then for all other 14 electrodes (Fig. 4). The association between the change in F4 N100 amplitude from the baseline session to the follow-up session (follow-up F4 N100 amplitude minus baseline F4 N100 amplitude) and treatment outcome were also investigated using linear regression (Fig. 2). The change in mean F4 N100 amplitude from baseline to follow-up was evaluated using paired *t*-test (Supplementary Fig. 3). The difference in baseline F4 N100 amplitude between treatment responders and non-responders was assessed using the Mann Whitney *U* test (Fig. 3). The difference between the mean amplitude during the baseline period (used for baseline corrections) and pre-stimulation resting period was assessed using the Mann Whitney U test as well (Supplementary Fig. 7). The Bonferroni correction was used to adjust *p*-values for multiple comparisons (Bland and Altman, 1995). Where applicable, both the uncorrected p-value and corrected p-value are presented for ease of interpretation since the Bonferroni correction increases the risk of Type II errors (Garamszegi, 2006).

3. Results

3.1. Demographics

23 participants, 13 female and 10 male, between the ages of 26–61, were included in the study. The mean age in years was 41.8 ± 10.6 SD. Full demographic and clinical characteristics are given in Table 1. Average baseline BDI-II was 34.0 ± 9.3. The response rate 4 weeks after treatment was 39.1 %. The remission rate 4 weeks after treatment was 34.8 %. Full clinical outcomes are summarized in Table 2.

3.2. Association between N100 amplitude and treatment outcome

Baseline N100 amplitude at the F4 electrode was found to have a significant positive association with treatment outcome ($p < .001$), suggesting that larger N100 amplitude at baseline led to greater improvement in BDI score. Every decrease in μV for F4 N100 amplitude

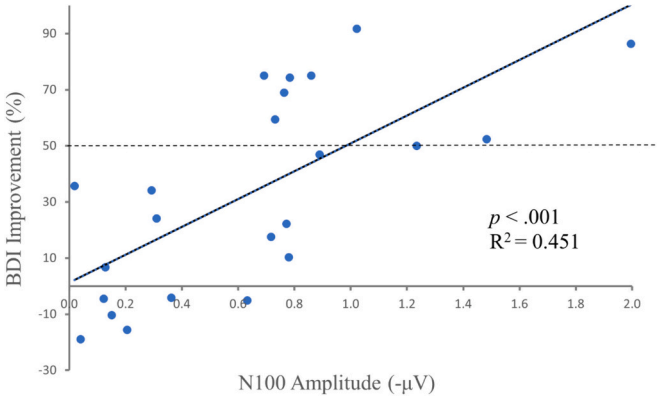


Fig. 1. Association between baseline F4 N100 amplitude and BDI-II percent improvement.
Baseline N100 amplitude at the F4 electrode was found to have a significant positive association with treatment outcome ($p < .001$), suggesting that larger N100 amplitude at baseline led to greater improvement in BDI-II score. Every decrease in μV for F4 N100 amplitude led to a 49.61 % increase in BDI-II percent improvement (slope = 49.61) and 45.1 % of the variance in BDI-II percent improvement was explained by the model containing only the baseline F4 N100 amplitude as the independent variable ($R^2 = 0.451$).

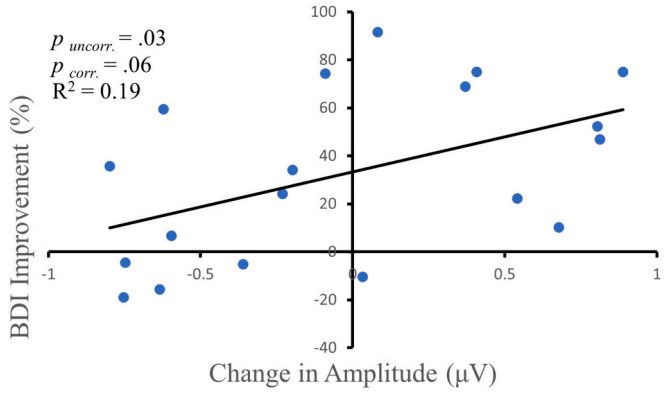


Fig. 2. Association between the change in F4 N100 amplitude and BDI-II percent improvement. The change in F4 N100 amplitude from the baseline session to the follow-up session (follow-up F4 N100 amplitude minus baseline F4 N100 amplitude) was significantly positively associated with treatment outcome prior to Bonferroni correction (p = .03 uncorrected, p = .06 corrected), suggesting that patients with larger decrease in F4 N100 amplitude post-treatment (more positive change in amplitude) experienced more improvements. Every μV increase in the change of F4 N100 amplitude results in a 29.2 % increase in BDI-II percent improvement (slope = 29.2) and 19 % of the variance in BDI-II percent improvement was explained by the model containing only the change in F4 N100 amplitude as the independent variable (R^2 = 0.19).

(decrease in μV equates to increase in N100 amplitude, since N100 is a negative peak) led to a 49.61 % increase in BDI-II percent improvement (slope = 49.61) and 45.1 % of the variance in BDI-II percent improvement was explained by the model containing only the baseline F4 N100 amplitude as the independent variable (R^2 = 0.451) (Fig. 1). This analysis was repeated after removing non-standard TEPs and the association remained significant despite decreased sample size (Supplementary Fig. 1). TEPs for individual patients are presented in Supplementary Fig. 2a and b. The change in F4 N100 amplitude from the baseline session to the follow-up session (follow-up F4 N100 amplitude minus baseline F4 N100 amplitude) was significantly positively associated with treatment outcome prior to Bonferroni correction (p = .03 uncorrected, p = .06 corrected), suggesting that patients with larger decrease in F4 N100 amplitude post-treatment (more positive change in amplitude) experienced more improvements. Every μV increase in the change of F4 N100 amplitude results in a 29.2 % increase in BDI-II percent improvement (slope = 29.2) and 19 % of the variance in BDI-II percent improvement was explained by the model containing only the change in F4 N100 amplitude as the independent variable (R^2 = 0.19) (Fig. 2). The mean F4 N100 amplitude decreased from baseline (−0.66 ± 0.50 μV) to follow-up (−0.58 ± 0.38 μV), however, the

Table 1
Demographic and clinical characteristics (n = 23).

Characteristics	No. (St. dev)
Age, years	41.8 (10.6)
Women	56.50 %
Age of onset, years	20.2 (10.0)
Length of current depressive episode, months	12.2 (12.3)
Baseline BDI-II	34.0 (9.3)
Antidepressant	57.9 %
Antidepressant combination	15.8 %
Antipsychotic augmentation	10.5 %
Psychostimulant augmentation	26.3 %
Benzodiazepine	5.3 %
ATHF total score	6.7 (5.4)
ATHF number of trials, current episode	1.3 (1.4)
ATHF highest score	3.2 (1.3)

Abbreviations: BDI-II = Beck Depression Inventory-II, ATHF = Antidepressant Treatment History Form.

Table 2
Clinical outcomes of 1 Hz R-DLPFC arTMS (n = 23).

Outcome measure: BDI-II	No. (St. dev)
Score baseline	34.0 (9.3)
Score 4 weeks after treatment	22.7 (14.2)
Percent improvement 4 weeks after treatment	33.6 % (35.2 %)
Response rate (>50 % improvement) 4 weeks after treatment	39.1 % (9/23)
Remission rate (≤12 on BDI-II) 4 weeks after treatment	34.8 % (8/23)

Abbreviations: BDI-II = Beck Depression Inventory-II.

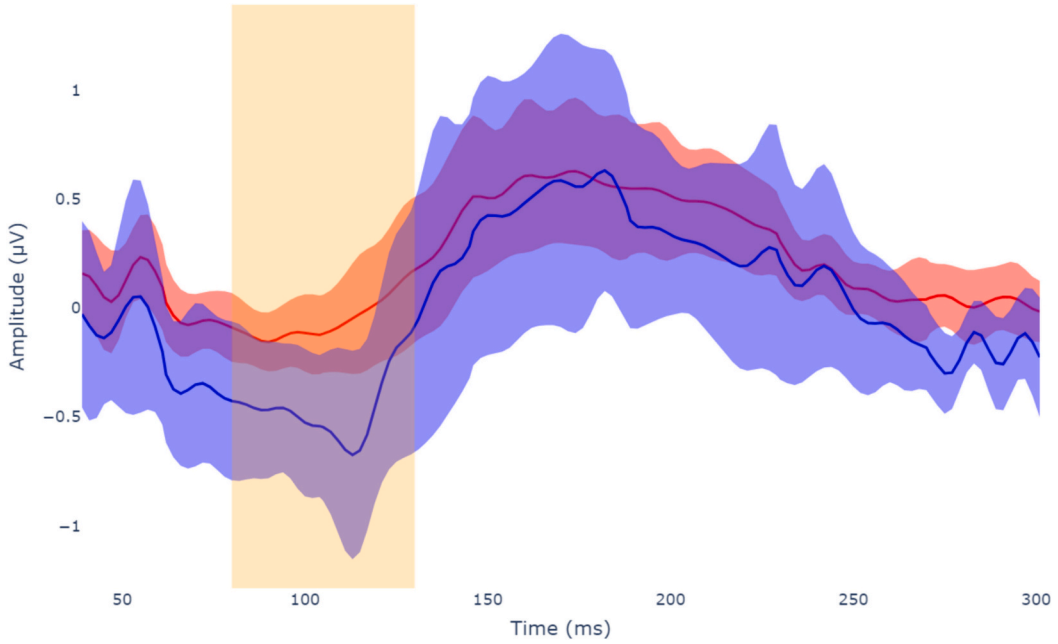


Fig. 3. Difference in baseline F4 TEPs between responders and non-responders. Responders had a significantly more negative mean baseline F4 TEP amplitude (−0.46 ± 0.15 μV) during the N100 time frame of 80 to 130 ms (vertical orange shading) compared to non-responders (−0.06 ± 0.09 μV) (p < .001, 95 % CI [0.35 0.47]). Line shadings represent 95 % confidence intervals.

decrease was not statistically significant ($t = -0.1603$, $p = .87$) (Supplementary Fig. 3). In addition, after rTMS treatment, patients with a lower baseline F4 N100 amplitude showed a larger increase (more negative change) in N100 amplitude whereas patients with a higher baseline F4 N100 amplitude showed a larger decrease (more positive change) in N100 amplitude ($p = .0001$, R2 = 0.5751); this suggests that the rTMS treatment may have a normalizing effect on N100 amplitude in MDD patients (Supplementary Fig. 8). Supplementary Fig. 4 demonstrates the average baseline F4 TEP for all patients, with the shaded region representing the timeframe of N100. Fig. 3 demonstrates the difference in mean baseline F4 TEP amplitude between treatment responders (BDI-II percent improvement ≥ 50) and non-responders (BDI-II percent improvement < 50). Responders had a significantly more negative mean baseline F4 TEP amplitude (-0.46 ± 0.15 μV) during the N100 time frame of 80 to 130 ms compared to non-responders (-0.06 ± 0.09 μV) ($p < .001$, 95 % CI [0.35, 0.47]).

The association between baseline N100 amplitude and BDI-II percent improvement for all electrodes is presented in Fig. 4. Significance is presented as −log of the uncorrected *p*-values. The baseline N100 amplitude is most significantly associated with BDI-II improvement at the R-DLPFC region (p at F4 < 0.001, p at F8 $= 0.028$) and the left-DLPFC region (p at F3 $= 0.015$). After Bonferroni correction, only the baseline F4 N100 amplitude remains significantly associated with BDI-II percent improvement (p $< .001$). A topographic map of t-values for the association between baseline N100 amplitude and treatment outcome is presented in Supplementary Fig. 5. Full statistical summary of the topographic plots is presented in Supplementary Table 1.

4. Discussion

To the best of our knowledge, this is the first study to assess the association between baseline N100 amplitude and treatment outcome of low frequency accelerated R-DLPFC arTMS in MDD. Our results, which were obtained through practical EEG and rTMS methods that do not require particularly extensive apparatus to be added to conventional clinical setups, are in line with previous reports and thus add to the growing body of evidence supporting the potential of utilizing baseline TEPs as biomarkers of rTMS treatment outcome in MDD. First and foremost, the baseline F4 N100 amplitude had a strong positive association with BDI percent improvement (Fig. 1). Second, prior to Bonferroni correction, the change in F4 N100 amplitude from baseline to follow-up had a weak positive association with treatment outcome (Fig. 2). Also, the mean F4 N100 amplitude decreased after arTMS treatment, but this decrease was not statistically significant (Supplementary Fig. 3). Moreover, the mean baseline F4 TEP amplitude during the N100 time frame was found to be significantly more negative for treatment responders than non-responders (Fig. 3). Lastly, prior to Bonferroni correction, baseline N100 amplitudes at the F3 electrode, representing the left-DLPFC (L-DLPFC) region, and F4 electrode, representing the R-DLPFC region, were found to be significantly associated with treatment outcome. Baseline N100 amplitudes at other electrodes had no statistically significant association with treatment outcome (Fig. 4).

The strong positive association between baseline F4 N100 amplitude and treatment outcome (Fig. 1), which suggests patients with larger F4 N100 amplitude at baseline led to larger improvements in MDD symptoms post 1 Hz arTMS, is in line with results by Sun and colleagues, who utilized MST as the treatment instead of 1 Hz rTMS and used suicidality scores instead of BDI scores; this association was highly significant in both studies ($p < .001$) (Sun et al., 2016). Since N100 amplitude can be increased by the application of GABABR agonist baclofen and is thus linked to slow cortical inhibition (Premoli et al., 2014), our result

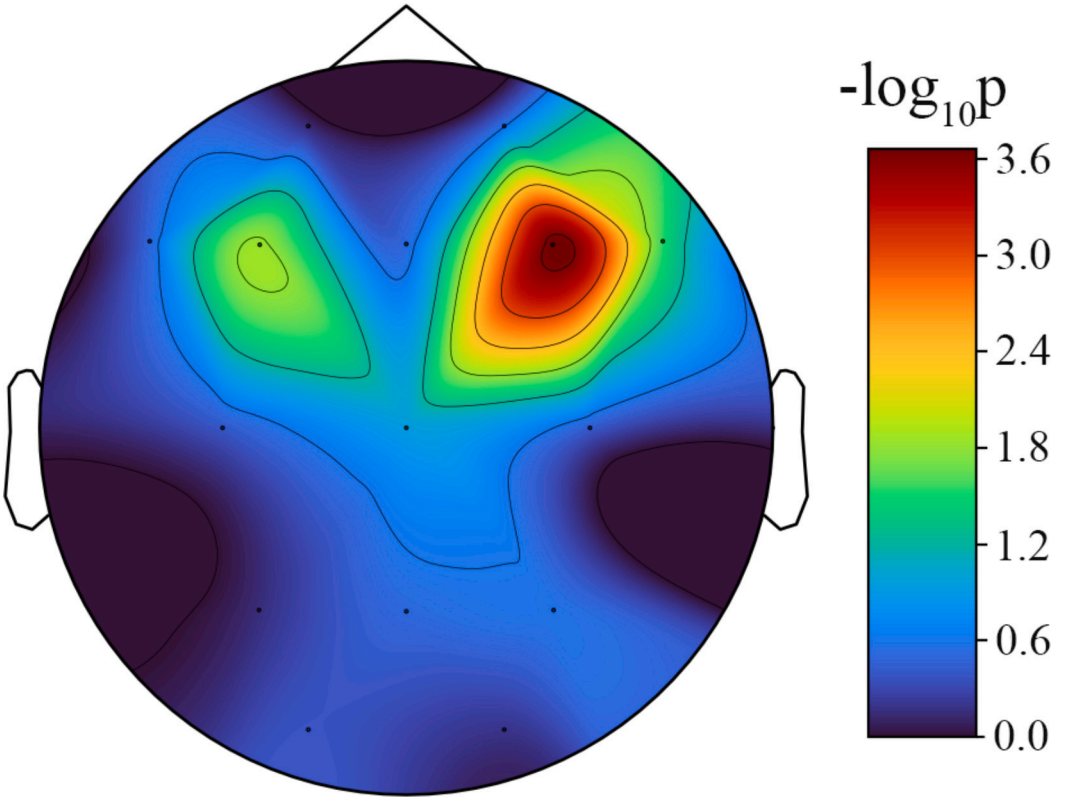


Fig. 4. Association between baseline N100 amplitude and BDI-II percent improvement for all electrodes. Significance is presented as −log of the uncorrected *p*-values. The baseline N100 amplitude is most significantly associated with BDI-II improvement at the R-DLPFC region (p at F4 < 0.001, p at F8 $= 0.028$) and the left-DLPFC region (p at F3 $= 0.015$). After Bonferroni correction, only the baseline F4 N100 amplitude remains significantly associated with BDI-II percent improvement (p $< .001$).

suggests that MDD patients with greater slow cortical inhibition at baseline benefit more from 1 Hz low frequency rTMS treatment. Moreover, GABA deficiency (Gold et al., 1980; Hasler et al., 2007; Price et al., 2009; Schür et al., 2016) as well as the decrease in DLPFC GABAergic interneuron size and density (Rajkowska et al., 2007) seen in MDD patients indicate that patients with greater baseline N100 amplitude may have inhibitory circuitry that are more akin to that of healthy individuals and are thus more easily influenced by the stimulations; patients with substantially damaged inhibitory circuitry may require more intensive and prolonged treatments.

The weak positive association (uncorrected) between the change in F4 N100 amplitude and treatment outcome (Fig. 2) is also in line with previous reports by Voineskos and colleagues. Since the N100 is a negative TEP, a larger change in F4 N100 amplitude (post-treatment N100 amplitude is less negative than baseline value) indicates a more sizable decrease in F4 N100 amplitude and suggests MDD patients who experience larger decreases in cortical inhibition after 1 Hz arTMS will see more improvements in MDD symptoms post treatment. A statistically non-significant decrease in mean F4 N100 amplitude for all patients post arTMS was also observed (Supplementary Fig. 3), which agrees with results from Voineskos and colleagues in trend (Voineskos et al., 2021). However, these results are in apparent contradiction with studies speculating that MDD patients are GABA deficient and that MDD treatments should thus increase cortical inhibition, which would manifest as increased mean N100 amplitude post-treatment, in order to carry out their anti-depressant effects. If increasing cortical inhibition is the main driving mechanism behind the anti-depressant effect of 1 Hz R-DLPFC arTMS, the association between the change in baseline N100 amplitude and treatment outcome should also be negative, suggesting that patients who experience a larger increase in baseline N100 amplitude post-treatment will receive greater benefit from the treatment. Together with the previously discussed concordance with MST-based results from Sun and colleagues, such results suggest it is possible that the predictive potential of N100 amplitude may not be specifically associated with the effect of 1 Hz stimulation, but rather with neural underpinnings of N100 that are related to the therapeutic effects of a broad range of non-invasive brain stimulation techniques, i.e., HF-rTMS, LF-rTMS, and MST; since the mechanistic differences (excitatory vs inhibitory etc.) between such interventions are still unclear, further investigation is required.

Furthermore, we demonstrated a strongly significant difference in mean baseline F4 TEP amplitude during the N100 time frame (80 to 130 ms) between treatment non-responders and responders, with responders having a significantly more negative mean baseline F4 TEP amplitude (Fig. 3). Although the confidence intervals are large due to the small sample sizes of each group (9 for responders and 14 for non-responders), this preliminary finding supports the classification of patients into 1 Hz arTMS responders and non-responders using baseline single electrode TEP amplitude during the N100 time frame (mean amplitude during 80–130 ms) in addition to baseline F4 N100 amplitude (most negative peak during 80–130 ms) alone. Future studies with larger sample sizes and standardized setups can therefore aim to define an appropriate classification voltage threshold that can be used to direct each patient to the most beneficial treatment in a timely manner.

Lastly, the all-electrode topographic map of significance (−log p) for the association between baseline N100 amplitude and treatment outcome is very similar to the topographic map of the same purpose reported by Sun and colleagues (Sun et al., 2016). For both studies, the association between baseline N100 amplitude and treatment outcome is strongest near the F4 electrode (R-DLPFC region) and followed by the F3 electrode (L-DLPFC region); the spectral color pattern of significance also forms an x-like pattern in both studies. For TMS-EEG, Sun and colleagues placed the rTMS coil near the L-DLPFC region (Sun et al., 2016), while the rTMS coil was placed over the R-DLPFC region in our study. For treatment, our target area was F4 while Sun and colleagues had the coil focused on Fz. Interestingly, despite having the stimulator coil placed at contralateral hemispheres during TMS-EEG as well as having differing treatment targets, the resulting topographic maps were very similar between the two studies. This similarity has several preliminary implications: first, the association between baseline N100 amplitude and treatment outcome is likely not the result of common TMS-EEG artefacts such as muscle, electrode noise (from coil contact, increased impedance, etc.), or auditory artefacts since the opposing coil placement between the two studies did not result in mirrored topographic maps. If the contribution of such artefacts to the association between baseline N100 amplitude and treatment outcome is substantial, the strongest effect would be expected to occur at the hemisphere where the rTMS coil is placed, as this would be the region where the most muscle movements, electrode noises, and loudest sounds (DLPFC positioning of the coil is close to the ipsilateral ear) would take place. Furthermore, this result confirms the well-known involvement of DLPFCs in MDD (Yadollahpour et al., 2016) since no other regions manifested notable significance. However, because the opposing coil placements during TMS-EEG did not produce notable variations in significance patterns, and that, as suggested by Du and colleagues, the N100 is possibly a non-regionally specific marker of response of the cortex to stimulations (Du et al., 2017), the particular target site of TMS single pulses used to generate the N100 peaks is likely not critical. Nonetheless, it should be noted that the significance at R-DLPFC is much higher than that of L-DLPFC in both studies, which is possibly due to intrinsic mechanistic differences between L-DLPFC and R-DLPFC in MDD (i.e., hyperactivity versus hypoactivity respectively (Yadollahpour et al., 2016)) that is unrelated to TMS-EEG or treatment targets.

Limitations of this preliminary study include: a small sample size, the lack of control group, and the absence of auditory masking. Although our sample size is comparable to that of similar studies (Rogasch et al., 2014; Sun et al., 2016; Voineskos et al., 2021), a larger sample size is necessary to overcome false positive and negative findings and allow for an appropriately accurate determination of a baseline N100 amplitude (or mean baseline TEP amplitude) classification threshold to distinguish between treatment responders and non-responders. Having a control group with placebo treatment could have further validated our results. In addition, the interval between each successive pulse is 1000 ms in our study, which is significantly shorter than the interval pulse interval of other studies (Rogasch et al., 2014). This short inter pulse interval could have allowed TEPs to contaminate the baseline period used in baseline corrections. However, we were able to demonstrate that there was no statistically significant difference ($p = .93$) between the mean amplitude of the baseline period (-0.0012 ± 0.22 μV) and the mean amplitude of the pre-stimulation resting period recorded before the onset of the first rTMS pulse (-0.0010 ± 0.24 μV). This suggests that the influence of TEPs on the recording is no longer significant at 700 ms post-pulse and that the inter pulse interval of 1000 ms is adequate for the establishment of the baseline period used in baseline corrections (Supplementary Fig. 7).

In terms of the lack of auditory masking, Rogasch and colleagues have demonstrated that the FastICA removal of the auditory component did not significantly affect the N100 amplitude on a group level (Rogasch et al., 2014). In addition, Poorganji and colleagues have shown that the N100 amplitude at the TMS target site electrode was not significantly different between subjects with auditory masking and those without auditory masking (Poorganji et al., 2023). Moreover, if the neurological auditory component, despite of being often recognized as a TMS-EEG artefact, significantly altered the N100 amplitude, and the altered N100 amplitude still showed an association with treatment outcome, then the auditory artefacts may in fact be an inseparable neurological component of the N100 treatment response biomarker. To further investigate the effects of the auditory component, we analyzed the association between baseline F4 N100 amplitude and treatment outcome both with and without the auditory component, and our results show that the auditory component, which was clearly identifiable in the majority of the patients, did not significantly affect the association

between baseline F4 N100 amplitude and treatment outcome (p with auditory component = 0.000089, R^2 = 0.504; p without auditory component = 0.000273, R^2 = 0.451) (Supplementary Fig. 6). However, it should be noted that because of technical restrictions, temporal electrodes such as T3 and T4 (used as ECG electrodes) were not available. Data from temporal electrodes would have better assisted in the examination of whether the auditory component was a confounding factor since they are best situated to record the auditory evoked potential. If the auditory component indeed contributes to a significant proportion of the relationship between F4 N100 amplitude and MDD treatment outcome, there is then a possibility of using auditory evoked potential measures independently as a treatment response biomarker, which could simplify the biomarker acquisition procedure even further compared to TEP setups. Thus, despite our best efforts, the impact of the auditory component cannot be fully excluded and remains a potential confounding factor to be examined by future studies using high-density EEG.

To conclude, the results of the present study suggest that, in agreement with previous literature, baseline N100 amplitude is a promising treatment outcome prediction biomarker for accelerated 1 Hz R-DLPFC rTMS in MDD at the group level. In particular, our study demonstrated that a strong association between baseline N100 amplitude and treatment outcome is obtainable through cost-effective and widely adoptable (TEP recordings can be done concurrently with 1 Hz rTMS treatment sessions) TMS-EEG methods that may not require the additional utilization of apparatus unique to standalone TEP recording sessions. Future studies should aim to increase sample size, further investigate the mechanism of N100 as well as the potential confounding effect of the auditory evoked potential, standardize TMS-EEG techniques, and eventually aim to obtain a reliable classification threshold that can streamline clinical flow by promptly directing patients to the rTMS protocol that is most effective for them as an individual.

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CRediT authorship contribution statement

Jack Z. Sheen: Writing – review & editing, Writing – original draft, Visualization, Software, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Frank Mazza:** Formal analysis. **Davide Momi:** Formal analysis. **Jean-Philippe Miron:** Writing – review & editing, Investigation. **Farrokh Mansouri:** Formal analysis. **Thomas Russell:** Data curation. **Ryan Zhou:** Data curation. **Molly Hyde:** Data curation. **Linsay Fox:** Data curation. **Helena Voetterl:** Data curation, Conceptualization. **Elie Bou Assi:** Formal analysis. **Zafiris J. Daskalakis:** Writing – review & editing. **Daniel M. Blumberger:** Writing – review & editing. **John D. Griffiths:** Writing – review & editing, Supervision, Formal analysis. **Jonathan Downar:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

JZS, **FMazza**, **DM**, **FMansouri**, **TR**, **RZ**, **MH**, **LF**, **HV**, **EBA** do not report any conflict of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jad.2024.07.131.

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